



# AUTO INSURANCE CHURN ANALYSIS

DATA ANALYTICS BOOTCAMP  
CAPSTONE PROJECT  
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# PROBLEM STATEMENT

**Customer churn is a significant challenge for auto insurance companies.**

Losing customers leads to a loss of premiums, directly affecting revenue, as auto insurance relies on recurring payments. When churn occurs, companies face higher marketing and acquisition costs, which are more expensive than retaining existing clients.

A high churn rate indicates low customer loyalty, dissatisfaction, and could damage the company's reputation and future growth prospects. Additionally, it signals that customers are easily attracted by competitors, forcing the company to constantly adjust its pricing and offerings.

Long-term customers provide stable income, and high churn means losing these valuable relationships, ultimately impacting overall profitability.

As a Data Analyst, I aim to provide insights and develop strategies to improve customer retention and reduce churn.

# DATASET EXPLORATION AND CLEANING

Dataset was downloaded from:

<https://www.kaggle.com/datasets/merishnasuwal/auto-insurance-churn-analysis-dataset?select=address.csv>

- Joined the following csv files in tableau: address, customer, demographic & termination.
- Checked for null values, duplicates or incorrect entries.
- Worked out the date of report (2 Dec 2022) and used it to calculate the null values for Age in Year field.
- Converted the following measures to dimensions by using calculated fields: Home Owner, College Degree, Good Credit & Has Children.
- Added a calculated field for Service Status (active / terminated).
- Grouped the following segments by using calculated fields: Age, Income, Tenure, Current Annual Amount, Home Market Value.

Data Visualization in Tableau can be assessed via below link:

[https://public.tableau.com/views/Capstonetesting/ActivevsTerminatedPieChart?:language=en-GB&:sid=&:redirect=auth&:display\\_count=n&:origin=viz\\_share\\_link](https://public.tableau.com/views/Capstonetesting/ActivevsTerminatedPieChart?:language=en-GB&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link)

# META DATA

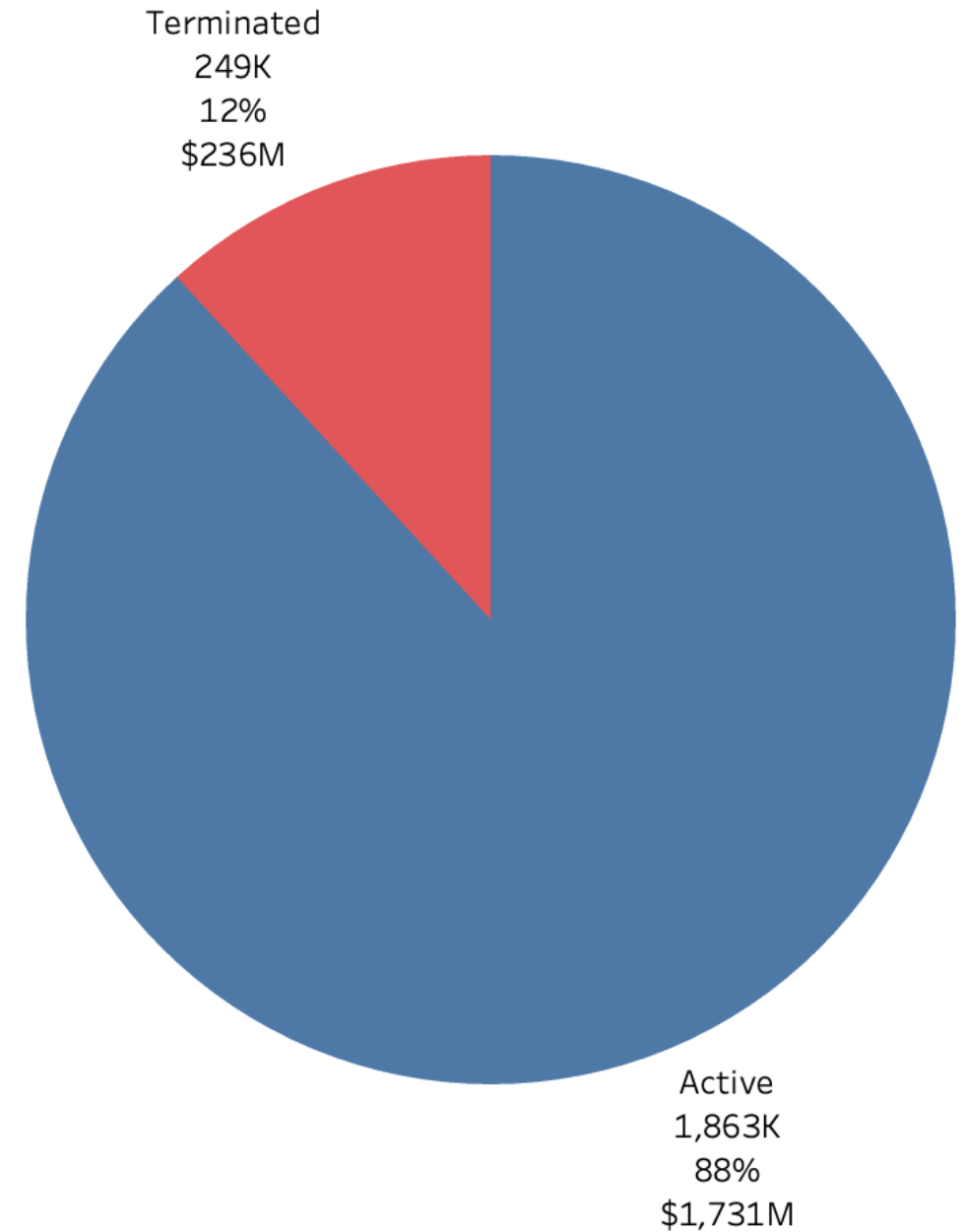
| Field Name      | Definition                                                                                                                                      |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| ACCT_SUSPD_DATE | Day of Account Suspension or Cancellation                                                                                                       |
| ADDRESS_ID      | Unique ID for the primary address associated with a customer                                                                                    |
| AGE_IN_YEARS    | Age of the individual.                                                                                                                          |
| CITY            | City                                                                                                                                            |
| COLLEGE_DEGREE  | Flag, 1 individual has a college degree or more, 0 otherwise.                                                                                   |
| COUNTY          | County                                                                                                                                          |
| CURR_ANN_AMT    | The Annual dollar value paid by the customer. It is not the policy amount.<br>It is actually amount the customer paid during the previous year. |
| CUST_ORIG_DATE  | The date the individual became a customer                                                                                                       |
| DATE_OF_BIRTH   | Individual's date of birth                                                                                                                      |
| DAYS_TENURE     | The time in days individual has been a customer with the insurance agency.                                                                      |
| GOOD_CREDIT     | Flag, 1 individual has FICO greater than 630, 0 otherwise.                                                                                      |

# META DATA

| Field Name             | Definition                                                                                                        |
|------------------------|-------------------------------------------------------------------------------------------------------------------|
| HAS_CHILDREN           | Flag, 1 indicates the individual has children in the home, 0 otherwise.                                           |
| HOME_MARKET_VALUE      | Estimate value of home.                                                                                           |
| HOME_OWNER             | Flag, 1 individual owns primary home, 0 otherwise.                                                                |
| INCOME                 | Estimated Income for the Household associated with the individual                                                 |
| INDIVIDUAL_ID          | Unique ID for a specific insurance customer                                                                       |
| LATITUDE               | Latitude of the address                                                                                           |
| LENGTH_OF_RESIDENCE    | Estimated number of years the individual has lived in their current home.                                         |
| LONGITUDE              | Longitude of the address                                                                                          |
| MARITAL_STATUS         | Estimated marital status. Married or Single.                                                                      |
| SOCIAL_SECURITY_NUMBER | Social Security Number. Note the middle two digits are XX to prevent any coincidental actual SSNs from appearing. |
| STATE                  | State                                                                                                             |
| STREET_ADDRESS         | Mailing Address of the address                                                                                    |

# ACTIVE VS TERMINATED

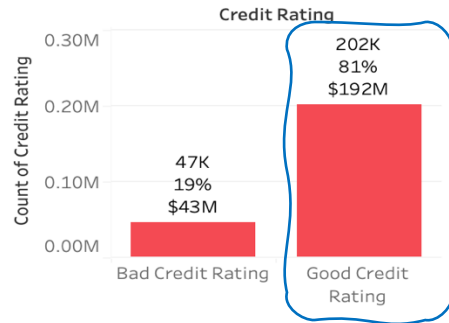
We will take a deep dive into the **Terminated** services to identify the customer segments with high churn risk, provide insights, and develop strategies to improve customer retention and reduce churn.



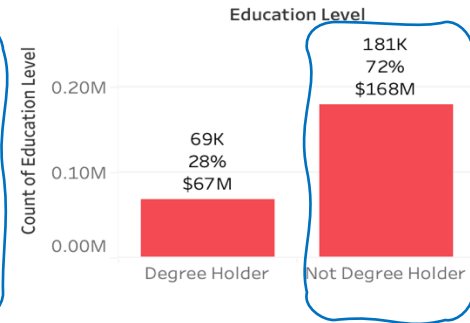
# IDENTIFYING HIGH CHURN SEGMENTS

Segments with significantly higher churn counts and churn amounts are identified.

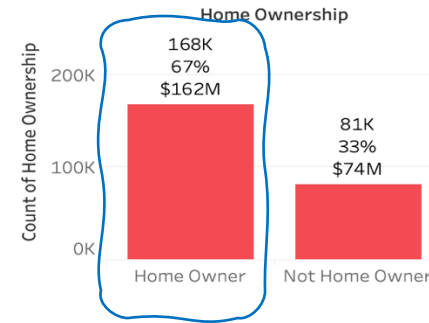
Credit Rating



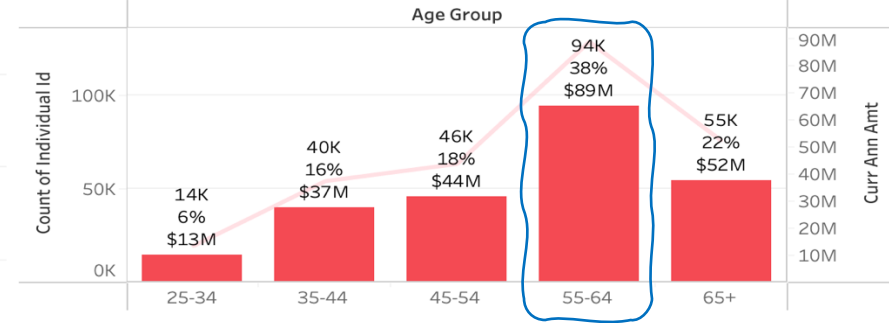
Education Level



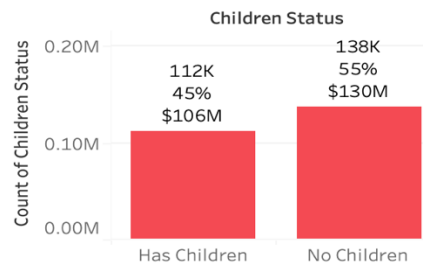
Home Ownership



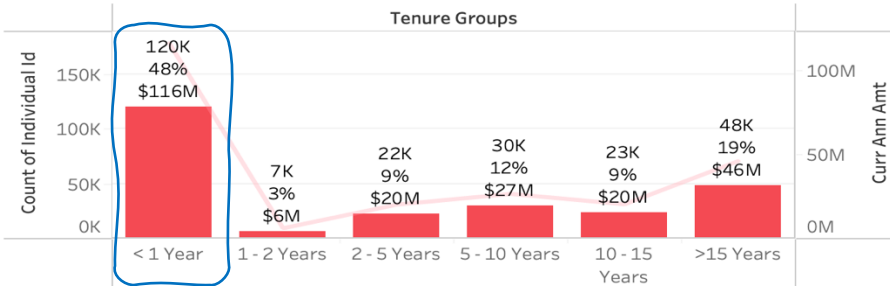
Age Group



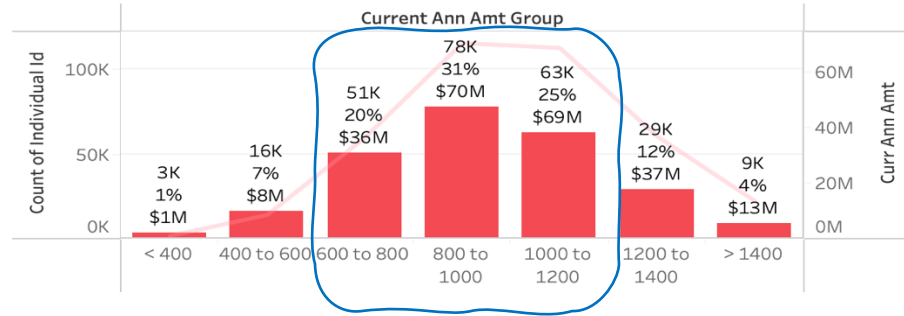
Children Status



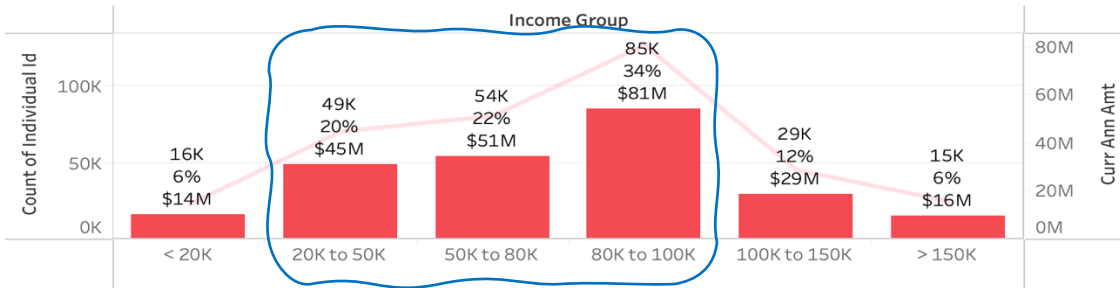
Tenure Group



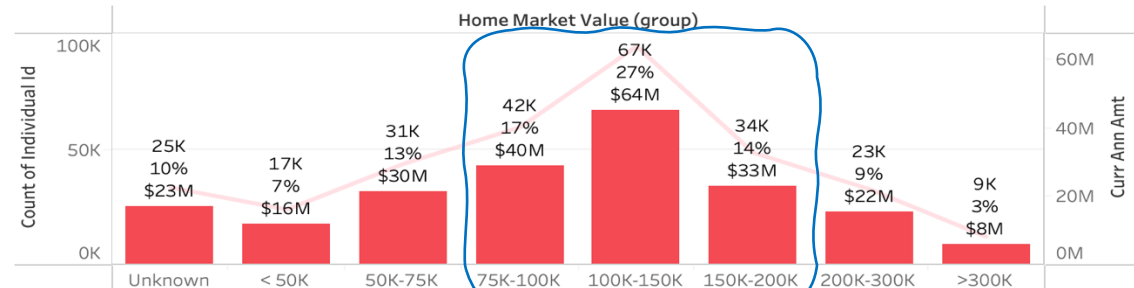
Current Ann Amt Group



Income Group

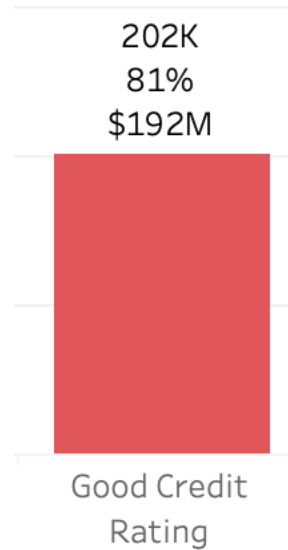


Home Value Group





# SEGMENT ANALYSIS



## Good Credit Rating

### Insights:

- Customers with good credit are **less risky** but also more **financially savvy**, meaning they shop around for better deals.
- They are more likely to **compare policies online** and leave if they find better rates.
- Higher churn here is alarming because these are **profitable, low-risk customers**.

### Strategies to Reduce Churn:

- ✓ Offer a **"Loyalty Discount"** for customers with a **good payment history**.
- ✓ Provide **personalized renewal pricing** based on tenure.
- ✓ Enhance **customer experience** with rewards for staying (e.g., cashback or deductible reductions).

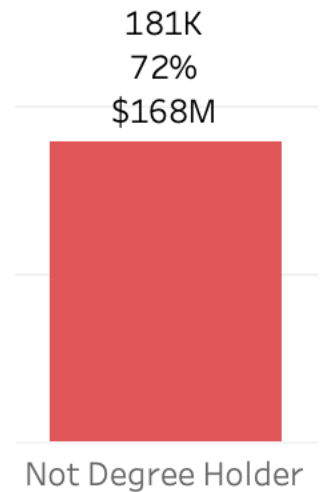
## Not Degree Holders

### Insights:

- Customers without a degree may **earn lower incomes**, making them **cost-conscious**.
- They could be more **price-sensitive** and prone to switching for small savings.
- They might have **less brand loyalty**, viewing insurance as a commodity.

### Strategies to Reduce Churn:

- ✓ Provide **flexible payment options** (e.g., installment plans).
- ✓ Target them with **value-driven messaging** instead of complex jargon.
- ✓ Offer **discounts for safe driving behavior** or **financial incentives** for long-term policyholders.





# SEGMENT ANALYSIS



## Homeowners

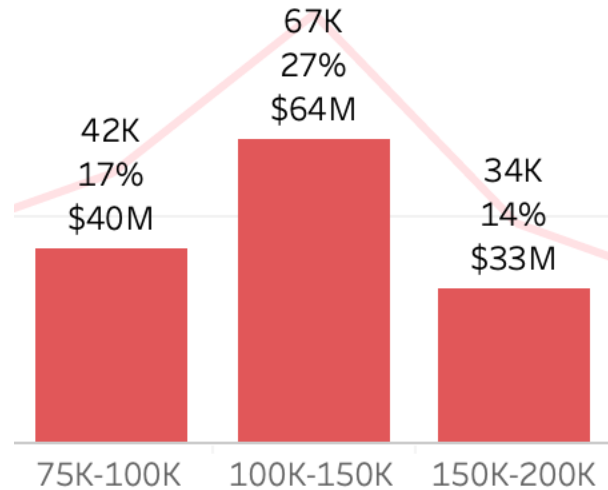
### Insights:

- Homeowners typically have **higher-value policies**, meaning their churn leads to **greater revenue loss**.
- They may be more **price-sensitive**, especially if premiums increase over time.
- They might switch to **bundle discounts** offered by competitors (e.g., home & auto insurance together).

### Strategies to Reduce Churn:

- ✓ Offer **multi-policy discounts** (bundling home & auto).
- ✓ Provide **loyalty perks** such as decreasing deductibles over time.
- ✓ Personalized engagement: Offer **home protection add-ons** to increase perceived value.

# SEGMENT ANALYSIS



## 💰 \$75K - \$100K Home Value

### Insights:

- These homeowners are **budget-conscious** and **highly price-sensitive**.
- They may prioritize **lower monthly costs over long-term savings**.
- More likely to **downgrade coverage or switch insurers for discounts**.

### Strategies to Reduce Churn:

- ✅ **Introduce Budget-Friendly Plans:** Offer **higher deductible options** to lower monthly premiums.
- ✅ **Flexible Payment Plans:** Allow installment-based payments to reduce short-term financial pressure.
- ✅ **Targeted Discounts:** Retention campaigns offering **limited-time premium discounts** to loyal customers.

## 💰 \$100K - \$150K Home Value

### Insights:

- **Highest churn risk** in this home value range.
- These customers are likely **mid-income earners** facing financial pressure.
- **Home & auto premium hikes** are a major churn factor.
- May be considering **switching to a competitor's bundled offer**.

### Strategies to Reduce Churn:

- ✅ **Competitive Price Matching:** Provide discounts when a competitor offers a lower price.
- ✅ **Home-Auto Bundle Promotions:** Incentivize customers to **stay by bundling home and auto insurance**.
- ✅ **Personalized Retention Offers:** Analyze customer **tenure & claims history** to offer tailored discounts.

## 💰 \$150K - \$200K Home Value

### Insights:

- **More financially stable** than lower-tier groups.
- Many are **loyal customers**, but still at risk due to **rising costs**.
- **Less likely to churn immediately**, but may explore options if rates keep increasing.

### Strategies to Reduce Churn:

- ✅ **Exclusive Loyalty Rewards:** Offer benefits like **free home inspections, premium protection, or smart home discounts**.
- ✅ **Proactive Retention Outreach:** Notify customers **before renewal** about available perks & pricing options.
- ✅ **Cost vs. Value Transparency:** Educate customers on **long-term cost savings by staying with the insurer**.

# SEGMENT ANALYSIS



## Tenure < 1 Year

### Insights:

- New customers may **leave quickly** if they experience poor service or find a cheaper alternative.
- Initial dissatisfaction (e.g., claim issues, billing confusion) can drive early churn.
- Some might be **temporary policyholders** (e.g., short-term rentals, relocating).

### Strategies to Reduce Churn:

- ✓ Implement **stronger onboarding programs** (clear guidance on claims, benefits, and renewal).
- ✓ Offer **first-year discounts or retention bonuses**.
- ✓ Improve **customer service response times**, ensuring early interactions are smooth.

## Age Group 55-64

### Insights:

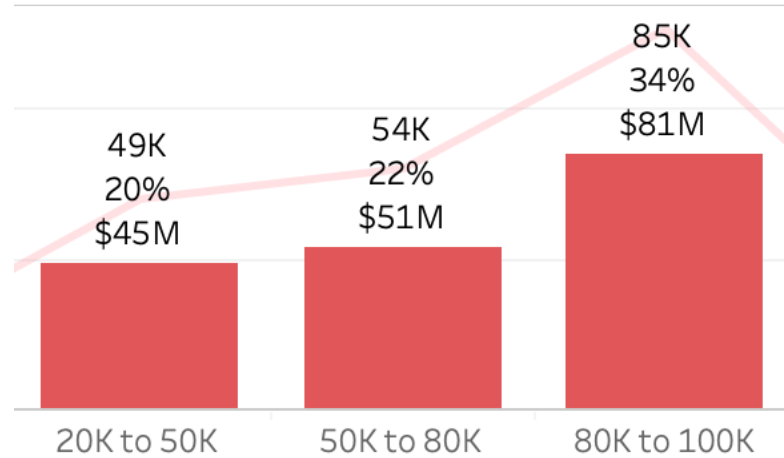
- Older customers may be **retiring and downsizing**, leading to fewer auto insurance needs.
- They may qualify for **senior discounts** from competitors.
- They might be looking for **simpler, more affordable policies**.

### Strategies to Reduce Churn:

- ✓ Introduce **senior-friendly discounts** to match competitor pricing.
- ✓ Offer **low-mileage discounts** (as older drivers may drive less).
- ✓ Highlight **customer service & ease of claims processing**, which seniors value.



# SEGMENT ANALYSIS



## 💰 \$20K - \$50K (Lower Income, Budget-Conscious)

◆ **Insight:** Customers in this segment are highly **price-sensitive** and may switch due to affordability concerns.

### ✅ **Retention Strategies:**

- Offer **budget-friendly policy tiers** with essential coverage at affordable rates.
- Provide **flexible payment plans** (monthly vs. annual payments).
- Implement **early renewal incentives** (small discounts for renewing before expiration).

## 💰 \$50K - \$80K (Middle Income, Value-Oriented Customers)

◆ **Insight:** These customers are **value-driven**—they are willing to pay for good service but still compare alternatives.

### ✅ **Retention Strategies:**

- Introduce **value-added services** (e.g., accident forgiveness, roadside assistance).
- Offer **transparent price breakdowns** showing long-term cost benefits.
- Leverage **personalized customer engagement** (e.g., tailored policy recommendations based on their needs).

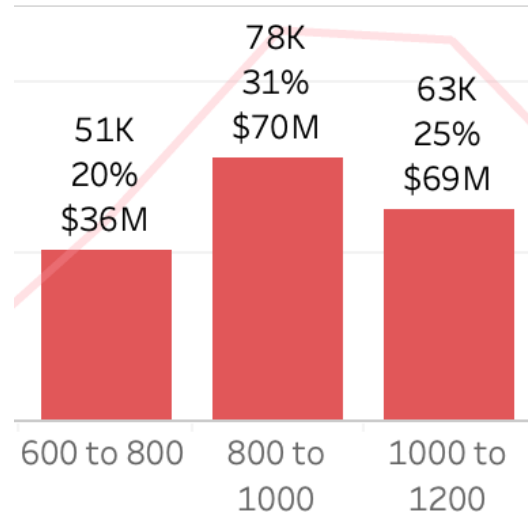
## 💰 \$80K - \$100K (Upper-Middle Class, Highest Churn Rate)

◆ **Insight:** This segment is **financially flexible** but also the most **aggressively targeted by competitors** offering discounts.

### ✅ **Retention Strategies:**

- Provide **customized loyalty discounts** based on tenure, claims-free history, and bundling.
- Enhance **digital experience** (seamless self-service, mobile app, 24/7 support).
- Offer premium 24/7 support and dedicated agents to engage them proactively.

# SEGMENT ANALYSIS



## **\$600 - \$800 Annual Amount**

### Insights:

- Customers in this bracket are still in the **affordable range** but may seek **better value for their money**.
- May feel **coverage is insufficient** and upgrade to competitors.
- **Financial struggles** might push them to cancel or downgrade.

### Strategies to Reduce Churn:

- ✅ **Upgrade Incentives:** Offer **discounts on higher coverage plans** to encourage retention.
- ✅ **Value Reinforcement:** Educate customers on the **risks of underinsurance**.

## **\$800 - \$1000 Annual Amount**

### Insights:

- These customers likely fall into the **mid-tier policy range**, feeling a **pricing pinch**.
- **Lack of perceived value** – They may expect more benefits at this price point.
- **Premium increases** might push them toward cheaper alternatives.

### Strategies to Reduce Churn:

- ✅ **Value-Added Benefits:** Provide perks like **free roadside assistance or claim-free rewards**.
- ✅ **Competitor Price Matching:** Offer **customized discounts** if a competitor has a lower premium.
- ✅ **Retention Alerts:** **Proactively engage** with customers before their renewal date.

## **\$1000 - \$1200 Annual Amount**

### Insights:

- Customers here **invest in higher coverage levels**, making them slightly more committed.
- **Affordability concerns** – Some may feel the coverage is too expensive.
- **Competitor promotions** – Other insurers may offer attractive discounts for high-value customers.
- **Service dissatisfaction** – At this premium level, customers expect superior service and benefits.

### Strategies to Reduce Churn:

- ✅ **Exclusive Loyalty Programs:** Offer **VIP customer support, premium discounts, or early renewal incentives**.
- ✅ **Flexible Deductible Options:** Allow policyholders to **adjust deductibles** to lower their premiums.
- ✅ **Risk-Based Discounts:** Reward **low-risk customers** with better pricing.



THANK YOU!